

DMLG Self-Governance: Empirical Evaluation of Modular Safety

Multi-Agent Moderation, Activation-Level Diversity, and Evaluator-Driven Control

Joachim De Witte

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Abstract

This paper presents the first empirical evaluation of *DMLG self-governance*: a multi-agent moderation architecture in which several independently trained DMLG evaluators govern the output of a writer agent through lemma-level ensemble voting. Unlike prior work, which introduced the Dynamic Modular Language Graph (DMLG) as a structural governance layer, the present study demonstrates that moderation is not merely a supporting capability but the *primary locus of reliability* in the architecture.

We show that a simple propose–vote–select mechanism, operating without beam search or additional training, is sufficient to stabilize generation, suppress drift, and enforce canonical structure. The moderation ensemble implements classical safety-engineering principles—redundancy, diversity, isolation, arbitration, and transparency—through architectural design rather than heuristics. Each moderator independently proposes admissible transitions; the ensemble selects only those lemmas supported by majority agreement, yielding a deterministic governance layer that is more reliable than any individual agent.

A key finding is that DMLG moderation can substantially improve the output of weak or under-trained writer agents. Even minimal-capacity writers produce coherent, grammatical, and canon-consistent text when governed by a diverse moderator ensemble. This establishes that *governance can dominate generation*: structural consensus among small evaluators can outperform the raw generative capacity of the underlying model.

The results position DMLG moderation as a lightweight, interpretable, and model-agnostic governance mechanism capable of stabilizing arbitrary generative agents. The architecture provides a practical path toward modular, reliable, and transparent language systems in which the writer generates, the ensemble governs, and the system remains stable.

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1 Introduction

Large language models (LLMs) have achieved impressive generative performance, yet their internal mechanisms remain opaque, statistically entangled, and difficult to control. Safety, coherence, and canonical style are typically enforced through post-hoc heuristics—such as RLHF, classifier-based filtering, or constrained decoding—rather than through intrinsic architectural guarantees. As a result, modern generative systems exhibit a structural asymmetry: generation is powerful but uncontrolled, while moderation is weak, heuristic, and external to the model.

The Dynamic Modular Language Graph (DMLG) proposes a fundamentally different perspective. Instead of treating moderation as an auxiliary correction mechanism, DMLG treats *governance* as the primary architectural role. A DMLG is a deterministic, interpretable graph whose nodes represent canonical tokens and whose edges encode admissible transitions. Small, modular evaluators—each with its own inductive bias—score transitions locally. In this framework, generation is merely a special case of graph traversal, while moderation becomes a first-class, structurally grounded operation.

This paper presents the first empirical evaluation of *DMLG self-governance*: a multi-agent moderation architecture in which several independently trained DMLG evaluators vote on each proposed lemma. The system implements ensemble arbitration through a simple propose-vote-select rule, combined with activation-diverse grammar consensus. Despite its simplicity, the mechanism yields a surprising and conceptually important result: **DMLG is empirically stronger as a moderator than as a generator**. Even extremely weak writer agents—such as HardTanh models with minimal expressive capacity—produce coherent, grammatical, and canon-consistent text when governed by a small ensemble of DMLG moderators.

This finding suggests that linguistic quality, safety, and canonical fidelity do not require a powerful generator. Instead, they emerge from the interaction of multiple small evaluators with diverse activation functions and inductive biases. Moderation, not generation, becomes the primary locus of reliability. This stands in contrast to transformer architectures, where generation is monolithic and moderation is heuristic.

The contributions of this paper are threefold:

1. We formalize DMLG self-governance as a modular, interpretable, activation-diverse moderation architecture.
2. We empirically demonstrate that multi-agent DMLG moderation can dramatically improve the output of weak or unstable writer agents, establishing that governance can dominate generation.
3. We show that consensus-driven moderation yields stable semantics, consistent lemma selection, and strong drift resistance, even when the underlying writer is noisy, under-trained, or stylistically divergent.

Together, these results position DMLG not as an alternative to LLMs, but as a *structural governance layer* capable of enforcing canonical style, traceable decision paths, and deterministic constraints in hybrid generative systems. The writer generates; the ensemble governs; and the system remains stable.

2 Motivation

Modern language systems are dominated by transformer-based LLMs, whose strengths and weaknesses are tightly coupled to their statistical foundations. They excel at broad generalization and fluent generation, but their internal mechanisms are opaque, high-dimensional, and difficult to constrain. As a result, current governance techniques—RLHF, classifier-based

filtering, prompt engineering, and constrained decoding—operate outside the generative mechanism and cannot provide deterministic guarantees about canonical structure, traceability, or domain-locked behaviour.

The Dynamic Modular Language Graph (DMLG) occupies a fundamentally different region of the design space. Its properties are orthogonal to those of mainstream architectures: deterministic tokens instead of embeddings, explicit graphs instead of latent spaces, modular evaluators instead of monolithic attention blocks, and PRE-based reasoning loops instead of stochastic sampling. This orthogonality makes DMLG far more suitable for *moderation* than for standalone generation.

Table 1 summarizes these contrasts. Where transformers prioritize expressive capacity, DMLG prioritizes structure, interpretability, and controllability. Where LLMs require massive datasets and GPU clusters, DMLG operates on small, domain-specific corpora and can run on commodity hardware. Where LLMs provide probabilistic fluency, DMLG provides deterministic constraints. These differences are not incremental—they represent a structural gap in the current AI ecosystem.

This gap is particularly evident in governance. No existing architecture offers:

- deterministic transition constraints,
- transparent token-level decision paths,
- ensemble-based arbitration with architectural diversity,
- canonical style enforcement independent of the writer model,
- or traceable drift detection at lemma resolution.

DMLG provides all of these by construction. Its explicit graph and modular evaluators make it possible to govern arbitrary generative agents—including LLMs—without modifying them. This enables applications such as company-approved writing style enforcement, domain-locked generation, and safety-critical traceability, which are difficult or impossible to achieve with statistical models alone.

Beyond scientific motivation, this represents a practical opportunity. As LLMs become commoditized, governance becomes a differentiating capability. A lightweight, interpretable, model-agnostic moderation layer that can be applied to any generative system provides a new axis of control and a new source of value. DMLG fills a structural lacuna in the field: it offers a principled, architecturally grounded solution to governance, rather than a heuristic patch on top of a black-box generator.

In this sense, DMLG is not an alternative to LLMs but a complementary architecture that supplies what they lack: deterministic structure, modular control, and transparent governance.

3 Related Work

3.1 Relation to Earlier Works by the Author

This paper extends a line of work on modular, interpretable architectures for language and reasoning.

A Dynamic and Modular Architecture for Language Modeling (DMLM) demonstrated that coherent language can emerge from modular, non-transformer components. However, that work focused on generation. In contrast, the present paper treats generation as secondary: the Dynamic Modular Language Graph (DMLG) is used primarily as a *governance layer*, not as a generative model.

Language as Graph Traversal: The DMLG Architecture introduced DMLG as a deterministic, canonical graph of tokens and transitions, with generation interpreted as traversal. That

Table 1: Comparison of DMLG with mainstream architectures.

Property	DMLG	Transformer	Mamba/SSM	RNN/RWKV
Complexity	Low, modular	Very high	Medium	Medium
Transparency	Full	Black box	Black box	Partial
Modularity	High	Low	Low	Low
Training cost	Low	Very high	High	Medium
Context scaling	Linear	Quadratic	Linear	Linear
Internal language	Canonical	Natural text	Natural text	Natural text
Embeddings	Deterministic	Trainable	Trainable	Trainable
Dataset needs	Small, domain-specific	Huge, mixed	Large	Large
Continual learning	Yes	No	Difficult	Difficult
Reasoning loop	PRE	Sampling/RLHF	Sampling	Sampling
Debuggability	Full	None	None	Limited
Hardware	CPU	GPU clusters	GPUs	GPUs

work proposed governance as a conceptual application but did not empirically evaluate moderation. The present paper fills this gap by demonstrating that DMLG moderation can dominate generation and stabilize arbitrary writer agents.

Adaptive Neural Networks explored custom activation functions and page-specific dynamics in modular systems. These ideas inform the design of DMLG’s local evaluators, which rely on small, interpretable components rather than large monolithic networks.

Across these works, the central theme is that explicit structure and modular evaluation can provide capabilities—traceability, controllability, and explainability—that are difficult to obtain from large-scale statistical models. The present paper provides the first empirical demonstration that these principles enable robust, ensemble-based governance.

3.2 Governance, Safety, and Traceability in Language Systems

Most research on safety and controllability in language models focuses on *post-hoc* mechanisms applied to transformer-based LLMs. These include:

- **Constrained decoding** Methods such as lexically constrained decoding and rule-based token filtering Hokamp and Liu 2017 restrict the model’s output space but operate outside the model’s internal structure.
- **RLHF and preference optimization** Reinforcement learning from human feedback Ouyang et al. 2022 and preference-based fine-tuning Bai et al. 2022 shape model behaviour but do not provide deterministic guarantees or traceable decision paths.
- **Safety layers and classifiers** External safety filters Ganguli et al. 2022 detect undesirable content but cannot enforce canonical style or structural consistency.
- **Explainability frameworks** Post-hoc interpretability methods Arrieta et al. 2020 provide partial insight into model behaviour but do not yield a structural model of allowed transitions.

These approaches share two limitations: (1) they operate *outside* the generative mechanism, and (2) they cannot provide deterministic guarantees about allowed transitions, canonical style, or domain fidelity.

DMLG moderation differs fundamentally. It provides:

- a *structural* representation of language (canonical graph),

- *deterministic* transition constraints,
- *ensemble-based arbitration* through lemma-level voting,
- *traceable decision paths* at token resolution,
- and *transparent disagreement signals* via vote counts.

Rather than modifying a generative model, DMLG governs it from the outside through a model-agnostic, deterministic structure.

3.3 Ensemble Methods and Safety-Critical Redundancy

Ensemble methods are widely used in machine learning for robustness and uncertainty estimation Lakshminarayanan, Pritzel, and Blundell 2017, but they typically combine probabilistic predictors rather than enforce structural constraints. Safety-critical systems, by contrast, rely on redundancy, diversity, isolation, and arbitration to prevent correlated failure modes.

DMLG moderation aligns with the latter tradition: multiple independently trained evaluators, each with distinct activation functions and inductive biases, jointly arbitrate the validity of each lemma transition. This yields a governance layer that is more reliable than any individual agent and provides deterministic guarantees unavailable in probabilistic ensembles.

3.4 Style Enforcement and Canonical Generation

Prior work on style transfer and canonical writing enforcement typically relies on fine-tuning, prompt engineering, or classifier-guided decoding. These methods can bias a model toward a target style but cannot guarantee structural fidelity or traceable decision-making.

DMLG moderation provides a different mechanism: style emerges from the canonical graph and the moderator ensemble’s internal canon. Lemma-level voting enforces stylistic and structural consistency without fine-tuning the writer model, enabling applications such as:

- company-approved writing style enforcement,
- domain-locked generation,
- canonical reproduction of narrative or technical styles,
- and transparent drift detection.

In this sense, DMLG moderation is closer to a *structural regulator* than a generative model or classifier.

4 Core Use Cases of DMLG Moderation

DMLG moderation enables four governance behaviours that arise directly from the interaction between a writer agent and a moderator ensemble. These behaviours require no additional training and follow solely from canonical graph structure, local evaluator proposals, and lemma-level voting. They illustrate how multi-agent moderation can constrain, guide, and analyse generative processes in a structurally transparent manner.

4.1 Steering Generation Through Ensemble Voting

At each generation step, the writer proposes a set of candidate lemma tokens. Each moderator independently evaluates these proposals by determining which lemmas it could also generate in the same canonical context. A token receives one vote per moderator that supports it. Tokens with the highest vote count form the eligible set, from which one is selected at random to preserve variation.

If no moderator supports any candidate, the system falls back to the writer’s top-ranked proposal. This ensures progress while maintaining canonical alignment whenever possible.

This ensemble-voting mechanism constrains traversal to regions of the canonical graph that are jointly supported by the moderator ensemble, providing a lightweight and deterministic form of generative steering.

4.2 Sentence-Level Confidence Scoring

Lemma-level voting yields a natural confidence measure for entire sentences. Given a sentence consisting of n lemma tokens and an ensemble of m moderators, the confidence score is defined as:

$$\text{confidence} = \frac{\sum_{\text{lemma } i} \text{votes}(i)}{n \times m}.$$

This score reflects the degree of ensemble agreement across the sentence and serves as a transparent acceptance criterion within the PRE (Propose–Retry–Evaluate) mechanism. Sentences with confidence below a chosen threshold may be rejected prior to grammar or semantic validation.

4.3 Token-Level Drift Visualization

Because each lemma token receives an explicit vote count, drift can be visualized directly in the natural-language output. Colour-coded annotations indicate the level of moderator agreement:

- **green** — full agreement,
- **orange** — partial agreement,
- **red** — no agreement.

This provides a fine-grained diagnostic view of where the writer diverges from the moderator ensemble, enabling transparent inspection of structural drift, stylistic inconsistency, or context misalignment.

4.4 Quality Stabilization for Arbitrary Writer Agents

Moderation can be applied to any writer agent, independent of architecture or training regime. The governance layer filters unstable transitions, suppresses structurally incompatible proposals, and guides the writer toward canon-consistent regions of the graph. This yields a general mechanism for stabilizing generative behaviour without modifying the writer itself.

4.5 Summary

These four behaviours—ensemble steering, confidence scoring, drift visualization, and writer stabilization—demonstrate that DMLG moderation constitutes a general-purpose governance mechanism. They arise naturally from canonical graph structure and evaluator design, without requiring additional training or model-specific adaptation.

4.6 The Niche Role of DMLG in the Modern AI Landscape

Large language models dominate contemporary NLP, but their strengths—expressive power, fluency, and broad generalization—come with structural limitations. Their internal representations are opaque, their decision paths are not traceable, and their outputs cannot be deterministically constrained to a canonical style or domain. Existing governance techniques—RLHF, safety classifiers, constrained decoding, and prompt engineering—operate outside the model’s internal structure and therefore cannot provide guarantees about allowed transitions, canonical fidelity, or stylistic consistency.

DMLG occupies a complementary niche. It is not a generative competitor to LLMs, but a *structural regulator* that provides:

- **deterministic transition constraints** through a canonical graph,
- **transparent decision paths** via lemma-level voting,
- **ensemble-based arbitration** that exposes uncertainty,
- **style and canon enforcement** independent of the writer model,
- **model-agnostic governance** applicable to any generative agent.

Where LLMs excel at producing content, DMLG excels at *governing* content. The architecture provides a mechanism for enforcing company-approved writing styles, domain-locked generation, canonical reproduction of narrative or technical styles, and traceable drift detection—capabilities that are difficult or impossible to obtain from statistical models alone.

In this sense, DMLG does not replace mainstream AI systems; it augments them. It provides the structural guarantees, transparency, and controllability that LLMs lack, enabling hybrid systems in which:

the writer generates, the ensemble governs, and the system remains stable.

4.7 The Generative Competence Requirement for Moderation

A central empirical finding of this work is that effective moderation requires the moderator to possess generative competence. A moderator cannot reliably evaluate or constrain a writer unless it can itself generate the same class of canonical transitions. This leads to the following structural principle.

Theorem 1 (Generative Competence Requirement). *A governance agent can only moderate a generative agent if it is itself capable of generating all transitions that it is expected to validate. Formally, for a writer W and moderator M , moderation is well-defined only if*

$$\mathcal{T}(W) \cap \mathcal{T}(M) \neq \emptyset,$$

and moderation is reliable only if

$$\mathcal{T}(M) \supseteq \mathcal{T}_{\text{valid}}(W),$$

where $\mathcal{T}(\cdot)$ denotes the set of admissible canonical transitions.

Interpretation. A moderator must be able to *generate* any transition it is expected to approve. Without this property, the moderator cannot distinguish between valid and invalid proposals, cannot detect drift, and cannot participate meaningfully in ensemble voting. Moderation therefore presupposes generative ability, even if the moderator is never used as a primary generator.

Implications. This principle explains several empirical observations:

- Moderators trained on archaic narrative corpora can govern writers trained on related archaic corpora, but not writers whose canonical structure is orthogonal to their own.
- Weak writers can be stabilized by strong moderators, but weak moderators cannot reliably govern strong writers.
- Ensemble diversity is beneficial only when each moderator retains sufficient generative competence to evaluate the writer’s proposals.

Conclusion. Moderation is not a purely discriminative task. It is a generative task performed in reverse: to judge whether a transition is valid, the moderator must be able to produce it. DMLG moderation therefore reveals a structural duality between generation and governance: the ability to generate canonical structure is a prerequisite for enforcing it.

5 The DMLG Moderation Algorithm

DMLG moderation is implemented as a lightweight, deterministic governance mechanism that operates on top of any writer agent. The algorithm requires no beam search, no additional training, and no model-specific adaptation. It relies solely on canonical graph structure, local evaluator proposals, and lemma-level ensemble voting. Despite its simplicity, the mechanism provides strong steering, drift suppression, and structural stabilization.

5.1 Overview

At each generation step, the writer proposes a set of candidate tokens. Each moderator independently proposes its own candidate set for the same canonical context. A token receives one vote per moderator that also proposes it. The token with the highest vote count is selected, subject to a grammar-consensus constraint. If no moderator supports any candidate, the system falls back to the writer’s top-ranked proposal.

This procedure yields a deterministic, model-agnostic governance layer that constrains traversal to regions jointly supported by the moderator ensemble.

5.2 Token Proposal

Let W be the writer agent and $M_1, \dots, M_k$ the moderator agents. Given a context state C , the writer produces a proposal set:

$$P_W = \{t_1, \dots, t_n\}.$$

Each moderator independently produces its own proposal set:

$$P_{M_j} = \{u_1, \dots, u_{m_j}\}.$$

No assumptions are made about the internal architecture of W or M_j ; the only requirement is that each agent can propose a finite set of canonical tokens for the given context.

5.3 Ensemble Voting

For each writer-proposed token $t \in P_W$, the vote count is defined as:

$$\text{votes}(t) = \left| \{j \in \{1, \dots, k\} : t \in P_{M_j}\} \right|.$$

Let

$$v_{\max} = \max_{t \in P_W} \text{votes}(t).$$

If $v_{\max} = 0$, the system selects the writer’s top-ranked token. This ensures forward progress even when the ensemble provides no guidance.

5.4 Grammar-Consensus Filtering

Grammar tokens require a minimum number of moderator votes. Let $\text{isLemma}(t)$ denote whether t is a lemma token. A token t is eligible if:

$$\text{eligible}(t) = \begin{cases} \text{votes}(t) = v_{\max}, & \text{if } \text{isLemma}(t), \\ \text{votes}(t) \geq \tau_{\text{grammar}}, & \text{otherwise,} \end{cases}$$

where τ_{grammar} is a fixed grammar-consensus threshold.

The final candidate set is:

$$E = \{t \in P_W : \text{eligible}(t)\}.$$

If $E = \emptyset$, the sentence is aborted and rejected by PRE.

If $E \neq \emptyset$, the next token is selected uniformly at random from E to preserve stylistic variety.

5.5 Sentence Construction and Termination

Tokens are appended sequentially until either:

- an end-of-line token is produced (successful termination), or
- the maximum token limit is reached (unsuccessful termination).

Unsuccessful sentences are rejected by PRE and do not enter the context state.

5.6 Confidence Scoring

For a generated sentence with lemma tokens $t_1, \dots, t_n$ and vote counts $v_1, \dots, v_n$, the confidence score is:

$$\text{confidence} = \frac{\sum_{i=1}^n v_i}{n \cdot k}.$$

This score is used as a deterministic acceptance criterion within PRE.

5.7 Properties

The algorithm has the following properties:

- **Model-agnostic:** applicable to any writer architecture.
- **Training-free:** moderators require no additional training.
- **Deterministic governance:** all decisions follow from canonical structure and ensemble agreement.
- **Low computational cost:** no beam search is required.
- **Stability:** grammar-consensus filtering suppresses drift.

Despite its simplicity, the algorithm provides strong generative steering and structural stabilization, and can outperform unguided beam-search decoding in practice.

6 Structural Analogy to Safety Engineering

The DMLG moderation ensemble exhibits the same structural principles that underlie classical safety-critical engineering systems. In such systems, the functional layer may be simple or even unreliable, but the governance layer must be robust, redundant, and architecturally diverse. The same pattern emerges in DMLG: the writer agent provides expressive variation, while the moderator ensemble ensures stability, constraint, and drift suppression.

6.1 Safety-Critical Design Principles

The moderation ensemble implements five canonical principles of safety engineering:

1. **Redundancy** — multiple independent evaluators assess each proposed transition. No single evaluator determines system behaviour.
2. **Diversity** — moderators may differ in architecture, activation functions, datasets, or inductive biases. This prevents shared failure modes and reduces the risk of correlated drift.
3. **Isolation** — the writer cannot influence the moderators. Information flow is strictly unidirectional: moderators evaluate proposals but do not receive gradients or updates from the writer.
4. **Arbitration** — majority voting ensures that no single evaluator can unilaterally approve an invalid transition. Ensemble agreement is required for acceptance.
5. **Transparency** — disagreement among moderators is directly observable through vote counts. This exposes uncertainty and enables deterministic rejection criteria.

These principles are not aesthetic design choices. They are structural requirements for any governance layer that must be more reliable than the system it governs. The writer may generate unstable or stylistically divergent proposals, but the ensemble ensures that only transitions consistent with the canonical graph and evaluator consensus are admitted.

6.2 Governance as the Primary Reliability Layer

The writer agent may be a singleton because creativity tolerates variation. Generative instability is acceptable as long as it is bounded by a reliable governance layer. In contrast, the moderation layer must be an ensemble: safety does not tolerate single points of failure.

By combining multiple independently trained evaluators, the DMLG moderation ensemble:

- avoids catastrophic attractors,
- suppresses pathological drift,
- prevents unilateral stylistic collapse,
- and ensures that no single model can dominate the generative process.

In this sense, DMLG does not merely borrow ideas from safety engineering; it *implements* them. The writer generates. The ensemble governs. The system remains stable.

7 Moderator Dataset and Agent Ensemble Architecture

This section describes the datasets and agent ensemble used to implement cross-canon DMLG moderation. The goal is not to maximize stylistic similarity between writer and moderators, but to ensure that the moderators possess a stable, internally coherent canon that can serve as an external governance reference. All moderator agents are trained on archaic narrative corpora with consistent syntax and strong thematic anchors, making deviations from their canon highly salient.

7.1 Moderator Canon: The Time Machine

The primary moderator canon is *The Time Machine* by H. G. Wells, an archaic narrative corpus with distinctive syntactic rhythm and stable lemma-grammar patterns. Its stylistic consistency and thematic coherence make it well suited for lemma-level drift detection. The text is sufficiently different from the writer corpora to provide external governance, yet not so orthogonal as to eliminate shared structural patterns.

The Wells corpus provides three essential properties for moderation:

1. **Stable syntax and rhythm** — enabling reliable detection of structural drift.
2. **Strong thematic anchors** — providing consistent semantic expectations across contexts.
3. **Moderate canon distance** — far enough from the writer corpora to expose deviations, but close enough to share grammatical structure.

7.2 Moderator Ensemble: Three Mature Agents

The governance layer consists of a three-agent ensemble, each trained to convergence on the Wells corpus. The agents differ in activation functions (SiLU, GELU, and a SiLU-GELU hybrid), ensuring architectural diversity and reducing the risk of correlated failure modes. All three agents operate solely as moderators; they do not generate primary text.

These agents provide:

- complementary inductive biases,
- sensitivity to syntactic and semantic drift,
- stable lemma distributions,
- robust consensus formation across activations.

Activation diversity is a structural requirement: moderators must not share the same architecture or activation law, otherwise correlated errors may propagate through the ensemble.

7.3 Weak Writer Agents (Not Moderators)

Two additional agents are trained on the same corpus but with significantly reduced training time and weaker activation functions (LeakyReLU and HardTanh). These agents are intentionally under-trained and serve exclusively as writer agents in evaluation experiments.

Their role is to test whether DMLG moderation can:

1. stabilize the output of weak or unstable writers,
2. suppress incoherent lemma proposals,
3. guide generation toward canon-consistent regions of the graph.

These agents are *not* part of the moderator ensemble.

7.4 Additional Archaic Writer Corpora

To evaluate cross-canon moderation beyond a single dataset, two additional archaic narrative corpora are used as writer sources:

Aesop’s Fables. A compact collection of moral tales with simple syntax and repetitive narrative patterns. Its regular structure makes it suitable for testing canonical tokenization and low-complexity world modelling.

Dr. Jekyll and Mr. Hyde. A Victorian novella with dense psychological themes and highly variable sentence structure. Its archaic style provides a challenging test for canonicalization and cross-canon moderation.

These corpora are stylistically distinct from the Wells canon but share enough structural features (archaic English, narrative prose, lemma–grammar patterns) to allow meaningful cross-canon evaluation.

7.5 Ensemble Dynamics

The three mature moderators play complementary roles:

- **Drift detection (SiLU, GELU).** Sensitive to deviations from the Wells canon, filtering anomalous lemma proposals.
- **Hybrid stability (SiLU+GELU).** Provides a stabilizing influence on borderline transitions by combining the inductive biases of both activations.
- **Consensus formation.** Lemma-level voting suppresses outliers, reinforces canon-consistent behaviour, and yields more stable transitions than any individual moderator.

Together, these agents form a structurally diverse ensemble capable of governing writers trained on both related and unrelated archaic corpora.

7.6 Dataset Selection and Public-Domain Constraints

All corpora used in this work are drawn from public-domain sources, primarily through Project Gutenberg. This choice is motivated by practical and scientific considerations: the *Triadic Cosmos* project aims to provide a fully reproducible and openly distributable reference implementation of DMLG moderation. Public-domain texts allow the entire training pipeline—including datasets, tokenization, and model checkpoints—to be shared without legal restrictions.

The selection of these corpora should therefore be understood as a constraint of open-science methodology rather than an attempt to identify stylistically optimal or domain-specific training material. Works such as *The Time Machine*, Aesop’s Fables, and *Dr. Jekyll and Mr. Hyde* are used because they are legally redistributable, structurally coherent, and sufficiently diverse to support cross-canon evaluation. Their inclusion reflects the practical requirement that all datasets in this study be accessible, inspectable, and reusable by the research community.

Importantly, the use of public-domain corpora highlights a central property of DMLG moderation: the architecture does not rely on carefully curated or proprietary datasets. Even when trained on openly available texts with heterogeneous styles and varying levels of structural regularity, the moderator ensemble exhibits robust drift detection, stable lemma-level voting, and consistent governance behaviour. This makes the approach suitable for open-source research environments where dataset constraints are both common and desirable.

Evaluation Methodology: LLM-Based Qualitative Assessment

All evaluations in this paper were performed by a large language model (LLM) operating as an external qualitative assessor. The evaluator examined coherence, structural stability, drift suppression, lemma-level consistency, and stylistic regularity across all generated samples. No numerical metrics were used; the analysis is based solely on observable structural properties and cross-sample consistency.

This approach ensures that the reported effects—such as emergent manifold selection, cross-canon stabilization, and governance-dominant behaviour—reflect qualitative differences that are robust enough to be detected by an independent model without access to the DMLG architecture or its internal mechanisms.

8 Architectural Diversity Through Activation Variation

Effective moderation requires that the evaluators in the ensemble do not share the same inductive biases or failure modes. In safety-critical systems, this principle is known as *architectural diversity*: redundant components must differ internally so that correlated errors do not propagate through the governance layer. In DMLG, this diversity arises naturally from differences in activation functions and, more importantly, from the use of learned activation MLPs.

8.1 Activation Functions as Inductive Biases

Each activation function imposes a distinct inductive bias on the evaluator’s transition preferences. Even when trained on the same canonical dataset, evaluators with different activations (e.g. SiLU, GELU, LeakyReLU, HardTanh) develop measurably different behaviours:

- different sensitivity to grammatical irregularities,
- different tolerance for semantic drift,
- different smoothness or sharpness in transition scoring,
- different stability under temperature variation.

These differences are not cosmetic. They ensure that moderators do not collapse onto the same decision manifold, enabling meaningful ensemble arbitration.

8.2 Learned Activation MLPs as Higher-Order Diversity

Beyond fixed activations, DMLG evaluators may employ *learned activation MLPs*—small neural networks that replace the activation function itself. Prior work on Adaptive Neural Networks has shown that such activation MLPs converge to dataset-specific nonlinearities that differ substantially across training runs, even when all other architectural components are identical.

This yields a stronger form of diversity:

- each evaluator learns its own activation geometry,
- activation MLPs encode corpus-specific “temperature laws,”
- evaluators diverge in how they score borderline transitions,
- ensemble disagreement becomes structurally meaningful.

Two evaluators trained on the same dataset but with different activation MLPs behave as if they were trained on different corpora. This makes them ideal moderators: they share the same canonical graph but disagree in precisely the regions where drift is most likely to occur.

8.3 Diversity as a Structural Requirement for Governance

Architectural diversity is not optional. It is a structural requirement for reliable moderation. If all moderators share the same activation law, the ensemble degenerates into a single model replicated k times, offering no protection against correlated drift or systematic errors.

Activation diversity ensures that:

- no single evaluator can dominate the ensemble,
- outlier transitions are suppressed by majority voting,
- disagreement exposes uncertainty and drift,
- consensus reflects structural validity rather than shared bias.

In this sense, activation variation—both fixed and learned—provides the architectural heterogeneity necessary for DMLG’s governance guarantees.

8.4 Conclusion

DMLG moderation relies on the interaction of multiple evaluators that differ in their activation functions, activation geometries, and inductive biases. This diversity is essential for ensemble arbitration, drift detection, and structural reliability. It ensures that the governance layer is more robust than any individual evaluator and that consensus reflects genuine canonical agreement rather than architectural homogeneity.

9 Quality Boosting Through Ensemble-Based Moderation

This section evaluates DMLG moderation as a mechanism for improving the output quality of weak and unstable writer agents. We compare three conditions: (1) unguided generation using beam search, (2) unmoderated sampling without beam search, and (3) sampling under DMLG ensemble moderation. The experiments use three writer agents of increasing strength—HardTanh, LeakyReLU, and a SiLU/GELU hybrid—and apply the same moderator ensemble (SiLU, GELU, and SiLU+GELU) across all conditions.

9.1 Unguided Generation: Attractor Collapse in Weak Writers

Weak writer agents exhibit severe attractor collapse when generating without moderation. The HardTanh writer produces long, repetitive sequences dominated by a small set of high-frequency motifs (e.g. “the greenery was ...”, “the figures ceased ...”, “I was only a match”), often repeating the same syntactic template dozens of times. The LeakyReLU writer shows a similar pattern, collapsing into motifs such as “the blackness”, “the workshop again”, and “the grindstone broke ...”. These attractors dominate the entire sequence, leading to semantic drift, syntactic instability, and loss of narrative structure.

Beam search amplifies this effect. Because weak writers have poorly shaped logit landscapes, beam search repeatedly selects the same local maxima, reinforcing the attractor and accelerating collapse. The resulting text is long, repetitive, and structurally unstable.

9.2 Moderation Without Beam Search: Stabilizing Weak Writers

When the same weak writers are sampled without beam search but under DMLG moderation, the output changes dramatically. The moderated LeakyReLU writer produces short, coherent, and syntactically well-formed sentences. Drift is visible through disagreement markers, but the

ensemble suppresses invalid or unstable transitions. Even extremely weak proposals are steered toward canon-consistent regions of the Wells corpus.

Moderation does not increase the expressive capacity of the writer; instead, it filters the writer’s proposals through a structurally diverse ensemble. Lemma-level voting removes attractor-driven repetitions and enforces local coherence. The result is text that is substantially more stable than the writer’s raw output, despite the absence of beam search or additional training.

9.3 HardTanh: Extreme Weakness and Partial Recovery

The HardTanh writer represents an extreme case: its activation function saturates aggressively, producing logits with minimal nuance. Unguided output is dominated by a handful of rigid templates and exhibits severe semantic erosion. Under moderation, the HardTanh writer still produces fragmented text, but the ensemble suppresses the worst degeneracies. Invalid transitions are filtered, repetitive loops are broken, and the output becomes locally coherent even if global structure remains weak. This demonstrates that moderation can partially recover quality even when the writer’s internal representation has collapsed.

9.4 Cross-Validation: Moderating a Strong Writer

To test whether moderation benefits only weak writers or also improves strong ones, we apply the same ensemble to a SiLU/GELU hybrid writer trained to convergence. Unguided output from this model is already substantially more coherent than that of the weak writers, but still exhibits attractor patterns (e.g. “the skin”, “a match”) and occasional syntactic drift.

Under moderation, these attractors are suppressed, and the output becomes more varied, more canon-consistent, and less prone to repetitive phrasing. The moderators do not simply replicate their own style: they filter the writer’s proposals while preserving its expressive capacity. This cross-validation confirms that moderation improves both weak and strong writers, though the magnitude of improvement is greatest for the weakest models.

9.5 Mechanism: Governance Dominates Generation

Across all experiments, a consistent pattern emerges: DMLG moderation provides a stronger source of reliability than the writer itself. Weak writers collapse into attractors; strong writers drift into stylistic loops; beam search amplifies both effects. The moderator ensemble, by contrast, enforces canonical structure through architectural diversity, lemma-level arbitration, and deterministic rejection of invalid transitions.

This leads to a counterintuitive but empirically supported conclusion: *governance can dominate generation*. Even minimal-capacity writers produce coherent, grammatical, and stylistically stable text when governed by a diverse ensemble of DMLG moderators. The ensemble does not need to be large; it needs to be diverse. Activation diversity, learned activation geometries, and independent training runs ensure that no single evaluator can dominate the decision process.

9.6 Conclusion

DMLG moderation provides a lightweight, interpretable, and model-agnostic mechanism for improving generative quality. It stabilizes weak writers, refines strong writers, and suppresses attractor collapse without beam search or additional training. These results demonstrate that the primary locus of reliability in a generative system need not be the generator itself: a small, diverse ensemble of evaluators can enforce canonical structure, suppress drift, and produce output that is more coherent than any individual agent.

10 Moderation as Canon Intersection

In addition to its role as a grammatical quality booster, moderation in our DMLG pipeline serves a second, equally important function: it enforces a *canon intersection* between a domain-specific generator and an external moderator ensemble. In this section we analyse this phenomenon using two contrasting literary domains: a trained Aesop-style fable generator and a trained *Jekyll & Hyde* generator. In both cases, the generator is moderated by the same three Time-Machine agents used in the quality-boosting experiments, but here moderation is applied *without* structural guidance or keyword prompting.

10.1 Experimental Conditions

For both Aesop and *Jekyll & Hyde*, we compare two configurations:

- **Fully Guided–Unmoderated:** A strong domain-specific writer receives explicit structural guidance, keyword constraints, and beam search. No moderation is applied.
- **Unguided–Moderated:** The same writer receives *no* structural guidance, no keywords, and no beam constraints. Instead, all generations are filtered by the three Time-Machine moderators, which enforce canonical consistency and reject incompatible continuations.

This setup isolates the effect of moderation itself: the generator is held constant, and only the presence or absence of the moderator ensemble changes.

10.2 Effect 1: Canon Intersection

Moderation does not generate text; it filters. The resulting output is therefore not a mixture but an *intersection* of two canons:

$$\text{Output Canon} = \text{Canon}_{\text{Writer}} \cap \text{Canon}_{\text{Moderators}}.$$

For Aesop, this intersection suppresses archaic or unstable constructions typical of unguided DMLG sampling and replaces them with the temporal, mechanical, and observational motifs characteristic of the Time-Machine moderators. The result is recognisably Aesopian in structure (animal agents, moral framing, short episodic scenes) but with a shifted semantic field: references to clocks, mechanisms, twilight, observation, and spatial orientation appear where classical Aesop would use rustic or pastoral imagery.

For *Jekyll & Hyde*, the effect is even more pronounced. The Hyde generator naturally produces dense Victorian diction, recursive introspection, and moralised psychological conflict. Moderation intersects this with the Time-Machine canon, yielding output that retains Hyde’s confessional tone and moral anxiety but replaces archaic Victorian syntax with more modern, temporally grounded phrasing. The result is a hybrid canon: recognisably Hyde, but filtered through the temporal and mechanical worldview of the moderators.

10.3 Effect 2: Grammatical and Structural Stabilisation

Across both domains, moderation also improves grammatical stability. Unguided DMLG sampling often produces:

- clause repetition,
- drifting referents,
- unstable verb tense cycles,

- local syntactic collapse under long sampling horizons.

The moderator ensemble rejects such continuations, forcing the generator to remain within a narrower, more syntactically coherent manifold. This stabilisation is not equivalent to style transfer: the generator’s sentence templates remain intact, but malformed or incoherent branches are pruned before they can propagate.

10.4 Summary

Moderation therefore has a dual effect:

1. **Canon Intersection:** The output becomes the intersection of the writer’s canon and the moderators’ canon, producing a hybrid but coherent literary style.
2. **Quality Stabilisation:** Moderation suppresses syntactic drift and enforces grammatical consistency, even without explicit structural guidance.

These results demonstrate that moderation is not merely a safety mechanism but a powerful tool for shaping the semantic and stylistic manifold of DMLG generators, especially on complex literary datasets such as Aesop and *Jekyll & Hyde*.

11 Moderation Using a Small Hybrid Modern Dataset

The previous section demonstrated that DMLG moderation remains effective even when the moderator ensemble and the writer are trained on distinct canonical corpora. We now extend this analysis to a more challenging setting: a small *hybrid modern dataset* consisting of three structurally incompatible regimes:

1. a simplified narrative corpus (short lemma-driven sentences with strong transition regularity),
2. a modern observational corpus (long-form prose centred on an observatory and caretaker),
3. and a poetic microfiction corpus (high-entropy, metaphorical, non-narrative text).

This dataset was intentionally constructed to contain conflicting inductive biases, incompatible syntactic rhythms, and heterogeneous transition structures. All writer and moderator agents were trained on this mixture.

11.1 Unguided Generation: Collapse Under Hybrid Training

When trained on the hybrid dataset, the writer agent exhibits severe instability. Even when decoded with beam search—the strongest unguided decoding strategy available to the model—the output shows:

- syntactic drift and broken clause boundaries,
- inconsistent subject–object transitions,
- uncontrolled mixing of narrative, observational, and poetic registers,
- and attractor collapse into repeated fragments

Beam search does not mitigate these effects; it amplifies them. Because the hybrid dataset contains no dominant structural attractor, the writer oscillates between incompatible regimes, producing text that is neither canonical nor coherent. This behaviour is consistent with the absence of a stable manifold in the underlying training distribution.

11.2 Moderation With Mix-Activation Evaluators

Despite the instability of the writer, a small ensemble of activation-diverse moderators trained on the same hybrid dataset produces *highly stable* moderated output. The ensemble consistently selects:

- short lemma-driven sentences,
- canonical DMLG-style transitions,
- and structurally coherent narrative fragments.

This is surprising because the hybrid dataset contains no explicit canonical core. The moderators nevertheless converge on a stable manifold that resembles the DMLG narrative regime, even though this regime constitutes only a fraction of the training data.

11.3 Emergent Manifold Selection

The key empirical finding is that moderation performs *manifold selection*. Among the three incompatible regimes in the hybrid dataset, only the DMLG-style narrative corpus exhibits sufficient structural redundancy to support reliable lemma-level voting. The poetic and observational regimes contain high syntactic entropy and low transition regularity, making them unsuitable for ensemble consensus.

As a result, the moderator ensemble implicitly identifies and stabilizes the only structurally coherent manifold available. This emergent behaviour occurs without explicit canonical supervision and demonstrates that DMLG moderation can extract a stable generative regime from heterogeneous data.

11.4 Beam Writer vs. Moderated Writer on a Distilled Dataset

To further test the generality of this effect, we evaluated a writer trained on a *distilled dataset* constructed from five heterogeneous corpora. The writer was decoded using both beam search and DMLG moderation with the same mix-activation ensemble.

LLM-based qualitative evaluation revealed that:

- beam search amplifies instability, producing long-range drift and semantic collapse,
- while DMLG moderation suppresses drift, enforces structural regularity, and yields coherent lemma-driven sequences.

This confirms that moderation dominates generation even when the writer is trained on a highly entropic distilled corpus. The ensemble consistently selects transitions that align with the emergent canonical manifold identified in the hybrid dataset experiments.

11.5 Conclusion

These results show that DMLG moderation is robust to heterogeneous training data. Even when both writer and moderators are trained on a mixture of incompatible styles, the ensemble identifies and stabilizes the only structurally coherent manifold available. This emergent behaviour suggests that DMLG moderation functions not merely as a constraint mechanism, but as a *structural filter* that extracts canonical order from statistical chaos.

The fact that all evaluations were performed by an LLM further underscores the qualitative clarity of the effect: the difference between unguided and moderated output is large enough to be consistently recognized without numerical scoring. This positions DMLG moderation as a general-purpose governance layer capable of stabilizing arbitrary generative agents, even in the absence of a well-defined canonical corpus.

12 Ensemble Size and the Role of Moderator Quality

All experiments in this paper were conducted with a three-moderator ensemble. This choice was not arbitrary: in practice, small ensembles tend to provide the most stable balance between diversity and consensus. Moderation relies on agreement between independent evaluators, and this agreement becomes harder to achieve as the ensemble grows and the canonical expectations of individual moderators diverge.

12.1 Quality Over Quantity

Our observations suggest that the *quality* and *alignment* of moderators matter more than the sheer number of them. A small set of well-trained, canonically coherent moderators provides:

- sufficient diversity to avoid overfitting to a single canonical view,
- enough overlap to reach consistent voting outcomes,
- low computational overhead,
- stable and interpretable moderation behaviour.

Larger ensembles introduce more canonical variation, which can be beneficial up to a point, but excessive diversity reduces the likelihood of consensus and may narrow the acceptance region to the point where generation becomes brittle or inefficient. This effect is architectural rather than empirical: moderation is a voting mechanism, and voting mechanisms naturally depend on the degree of shared structure among voters.

12.2 Practical Implications

These considerations motivate the use of small, high-quality moderator ensembles. In practice, a triad of moderators provides a good balance between independent judgement and shared canonical grounding, enabling stable moderation without excessive computational cost or consensus fragmentation.

This behaviour reinforces a broader theme of this work: moderation is most effective when it is *precisely calibrated*. The goal is not to maximise the number of moderators, but to ensure that the ensemble forms a coherent governance layer capable of constraining generation while preserving stylistic identity.

13 Limitations and Open Questions

Although the experiments in this paper show that DMLG moderation can substantially improve generative stability, canonical fidelity, and stylistic coherence, several limitations remain. These limitations do not reflect flaws in the architecture, but rather the broader challenge of combining structural governance with free-form generation.

Model Capacity. DMLG writers are intentionally lightweight and structurally constrained. Their generative weaknesses are a design choice: variation is delegated to the writer, while coherence is delegated to the moderators. As a result, unguided generation on complex corpora remains fragile without a governance layer. Moderation is therefore not optional but essential to the architecture.

Moderator Canon Dependence. Moderation enforces the canonical structure encoded in the moderator ensemble. When the moderator and writer canons differ, the output converges to their intersection. This behaviour is predictable and often desirable, but it also means that the choice of moderator canon directly shapes the resulting style. Selecting or training appropriate moderators remains an open design question.

Ensemble Size and Alignment. Moderation relies on agreement between independent evaluators. Our observations suggest that small, well-aligned ensembles provide the most stable behaviour: they offer enough diversity to avoid overfitting, yet enough overlap to reach consistent voting outcomes. Larger ensembles introduce more canonical variation, which can make consensus harder to achieve and may narrow the acceptance region. The challenge is therefore not to maximise ensemble size, but to calibrate it: moderator *quality* and *alignment* matter more than quantity.

Integration with Large Language Models. The most important open question concerns hybrid architectures. Transformers trained on large corpora are vastly stronger generators than DMLG writers; the question is not whether LLMs can generate, but whether they can *cooperate* with DMLG-style governance. In principle, moderation is agnostic to the source of candidate tokens, and the same acceptance–rejection mechanism could be applied to LLM sampling. However, several challenges remain:

- **Sampling compatibility:** LLM logits are dense and high-entropy, whereas DMLG moderation expects sparse, lemma-level transitions. Mapping between these representations requires further study.
- **Latency and throughput:** LLM sampling produces many candidate continuations per step; moderation must remain efficient enough to filter them in real time.
- **Canonical alignment:** LLMs tend to normalise stylistic irregularities, while DMLG moderators preserve them. Understanding how these tendencies interact is an open research direction.
- **Governance strength:** It is unknown how strongly a DMLG moderator can constrain a much more expressive generator without collapsing diversity.

Future Directions. These limitations point toward a broader research agenda: can DMLG moderation serve as a general-purpose governance layer for transformer-based generation? If so, hybrid systems may combine the expressive power of LLMs with the canonical stability and interpretability of DMLG. This paper provides initial evidence that moderation can dominate generation in lightweight settings; extending this to large-scale hybrid architectures remains an open and promising challenge.

14 Testable Predictions for LLM–DMLG Hybrid Moderation

The experiments in this paper evaluate DMLG moderation exclusively in the DMLG–DMLG setting: a lightweight writer governed by a small, aligned ensemble of DMLG moderators. Extending this mechanism to transformer-based generation raises several concrete, testable predictions. These predictions concern the behaviour of the moderation layer when applied to high-capacity LLM sampling, and can be evaluated without modifying the underlying architecture.

Prediction 1: Moderation Improves LLM Canon Fidelity. If an LLM is sampled freely on a stylistically rigid corpus (e.g. archaic literature), its output will tend to normalise the style into modern prose. Applying DMLG moderation to the LLM sampling process should increase canonical fidelity by rejecting continuations that deviate from the target style. *Test:* Compare LLM-only sampling with LLM+moderation on a fixed corpus and measure canonical deviation using a style classifier or perplexity under a domain-specific model.

Prediction 2: Moderation Reduces LLM Drift on Long Horizons. LLMs exhibit semantic drift and topic instability during long-form generation. DMLG moderation should reduce this drift by pruning off-canon continuations. *Test:* Generate long sequences with and without moderation and quantify topic drift using embedding-based coherence metrics.

Prediction 3: Moderation Narrows the LLM Sampling Distribution. Because moderation rejects high-entropy or stylistically inconsistent continuations, the effective sampling distribution of the LLM should become narrower and more structured. *Test:* Measure entropy of accepted vs. proposed LLM logits across many steps.

Prediction 4: Moderator Quality Dominates LLM Capacity. If moderation acts as a governance layer, then the quality and alignment of the moderators should matter more than the capacity of the LLM. A strong LLM with weak or misaligned moderators should perform worse than a moderate LLM with strong, canonically aligned moderators. *Test:* Cross LLMs of different sizes with moderator ensembles of varying quality and measure canonical fidelity.

Prediction 5: Small, Aligned Moderator Ensembles Outperform Large, Diverse Ones. The ensemble-size behaviour observed in DMLG–DMLG settings should generalise: a small, coherent moderator ensemble should outperform a large, heterogeneous one when governing LLM sampling. *Test:* Evaluate LLM+moderation with ensembles of size 1–N and measure acceptance rate, coherence, and canonical fidelity.

Prediction 6: LLM–DMLG Hybrid Moderation Preserves LLM Fluency. Because moderation filters rather than generates, it should preserve the fluency and grammatical competence of the LLM while constraining its style. *Test:* Compare fluency metrics (e.g. grammaticality, perplexity under a general language model) for LLM-only vs. LLM+moderation outputs.

Prediction 7: Moderation Enables LLMs to Generate in Styles They Do Not Natively Master. If moderation enforces canonical constraints independently of the generator, then an LLM should be able to produce stable output in a style it does not fully internalise, provided the moderators encode that style. *Test:* Use moderators trained on a niche or archaic corpus and evaluate whether LLM+moderation produces more canonical output than LLM-only sampling.

Prediction 8: Moderation Can Be Inserted Without Retraining the LLM. Because moderation operates at the sampling layer, it should function as a drop-in governance mechanism without requiring LLM fine-tuning. *Test:* Apply moderation to multiple off-the-shelf LLMs and measure improvements in canonical fidelity and drift reduction.

Prediction 9: Moderation Increases Interpretability of LLM Decisions. Moderator accept/reject patterns should reveal which stylistic or canonical constraints the LLM violates, providing a form of interpretable governance. *Test:* Log moderator decisions and cluster rejected continuations to identify systematic LLM deviations.

Prediction 10: Hybrid Moderation Exhibits the Same Failure Modes as DMLG Moderation. If moderation is architecture-agnostic, then LLM+moderation should show the same structural behaviours observed in DMLG–DMLG settings: narrow acceptance regions, dependence on moderator alignment, and sensitivity to ensemble size. *Test:* Replicate the ensemble-size and canonical-intersection experiments with LLM sampling.

15 Conclusion

This work demonstrates that DMLG moderation is not merely an auxiliary mechanism for filtering or stabilising generative output, but a structural component that fundamentally enhances the behaviour of the underlying architecture. Across all experiments—ranging from synthetic corpora to Aesop, *The Time Machine*, and *Jekyll & Hyde*—we observe the same pattern: *DMLG generates better text when it is governed by DMLG*.

Unguided writers, even when trained on rich literary corpora, exhibit well-known DMLG failure modes: attractor loops, syntactic fragmentation, semantic drift, and unstable clause structure. Moderation suppresses these degeneracies without introducing new content. Instead, the moderator ensemble acts as a canonical constraint operator: it accepts only those continuations that satisfy the structural, lexical, and stylistic invariants encoded in its own training corpus. The result is a form of *self-reinforcing governance*: the architecture improves its own generative behaviour not by increasing model capacity, but by constraining itself.

This effect is particularly striking on difficult corpora. On *The Time Machine*, where unguided DMLG generation collapses into unstable archaic fragments, moderation produces coherent, temporally grounded sentences that remain faithful to the canonical style. On *Jekyll & Hyde*, moderation restores psychological coherence and Victorian cadence while eliminating malformed recursive structures. Even on Aesop, moderation shifts the generator toward a stable intersection between the fable canon and the moderators’ temporal-mechanical canon, demonstrating that moderation can enforce stylistic identity even when the moderator and writer are trained on different corpora.

The central insight is therefore architectural: *DMLG is strongest not as a generator, but as a governor*. Its generative layer provides variation, but its moderation layer provides structure, coherence, and canonical fidelity. Where large language models avoid archaic or stylistically rigid corpora by normalising them into modern prose, DMLG preserves these canons by enforcing them. Moderation enables DMLG to operate reliably on corpora that would otherwise exceed its generative stability.

In this sense, DMLG offers a complementary paradigm to contemporary neural generation: a governance-first architecture in which quality, coherence, and stylistic identity emerge not from scale, but from constraint. The fact that DMLG becomes a better generator precisely when it moderates itself underscores its core contribution: *governance can dominate generation*.

Concluding Note

Transformers excel by learning broad linguistic priors; DMLG excels by enforcing narrow canonical structure. Where the former smooths irregular corpora into fluent modern prose, the latter preserves their internal logic through governance. Moderation therefore does not compete with generation—it completes it: a small DMLG writer becomes stronger precisely when it is constrained by its own evaluators. This complementarity suggests a hybrid future in which LLMs generate and DMLG governs, a separation of roles that may also prove useful in applied settings where consistency and traceability are required.

A Triadic Cosmos Ecosystem Context

This paper is part of the *Triadic Cosmos* ecosystem, which develops a unified conceptual and computational framework spanning physical ontology and modular architectures for intelligent systems. All materials, extended notes, and evolving documentation are available in the public repository: <https://github.com/triadic-cosmos>. A complete reference Python implementation is included in the ecosystem under the AGPL-3.0 license, together with example datasets and models provided in a separate data repository. These resources make the architecture fully inspectable, reproducible, and practically deployable.

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